

**TREATMENT AND EMERGENCY RECOMMENDATIONS FOR  
Carnitine Translocase Deficiency  
Carnitine Palmitoyl Transferase II Deficiency**

RE:

DOB:

MRN:

**THIS PATIENT HAS A RARE AND SERIOUS INHERITED DISORDER  
OF FATTY ACID METABOLISM AND IS AT RISK FOR  
COMPLICATIONS FROM THIS CONDITION.**

**Please see last section of this letter for  
Emergency Treatment during Decompensation**

Definition: These disorders are inherited problems of transport of long chain fatty acids (C14-C18) into mitochondria and/or the removal of carnitine from the fatty acids (C14, C16, C18) once in the mitochondria. The result in both cases is an inability to metabolize Very Long Chain Fatty Acids.

Signs of Decompensation: Lethargy, pallor, decreased state of consciousness, nonketotic hypoglycemia, elevated ammonia, elevated CK, seizures, coma, and sudden cardiopulmonary arrest.

Treatments:

A. ***Prevention of decompensation:***

1. Medium Chain Triglyceride formula (ie: Enfaport, Pregestimil)
2. Avoid fasting for long periods ("long periods" are relative and depend on age and size)
3. Supplement with L-carnitine 100mg/kg/24hrs (may be taken in 2, 3, or 4 doses during the day. L-carnitine (Carnitor, Trade name) comes in liquid form 100mg/ml, tablets 330mg, capsules 250mg.
4. Close follow-up and awareness of this condition by family and physicians

B. ***Treatments during an illness*** (colds, flu, ear infections, etc):

1. Maintain oral intake, may use glucose containing fluids if refusing formula
2. Continue oral Carnitine
3. If unable to maintain the two previous instructions or if lethargic, having persistent vomiting or decreased state of consciousness then use the EMERGENCY PROTOCOL

*Please see next page...*

- C. ***Treatment during decompensation*** (Lethargy, vomiting, decreased state of consciousness, low blood sugar, coma. Please note that not all of these symptoms need to be present):
1. Obtain laboratory studies BMP, CK, NH<sub>3</sub>
  2. Start IV fluids immediately: D10/ ½ normal saline 1½ times maintenance (add KCl as indicated).
    - i. Use D10 even if the blood sugar is normal, and for maintenance.
    - ii. Keep the blood glucose high 110-170mg/dL. Use D25 if necessary.
  3. IV carnitine (200 mg/ml).
    - i. Loading dose is 200 mg/kg of body weight given over 2 hours (L-carnitine is compatible with D10)
    - ii. To be followed by 50 mg/kg every 6 hours.
  4. Treat the underlying and precipitating illness (infection, etc), and obtain appropriate additional laboratory studies to establish the cause of the physiological challenge.
  5. **Notify the metabolic specialist immediately (after business hours this number allows you to speak with the geneticist on call:**  
**Kentucky Childrens Hospital**  
**Division of Genetics and Metabolism**  
R. Stephen Amato, MD, PhD  
859-323-0396  
Kentucky Clinic – J420  
740 S. Limestone Street  
Lexington, KY. 40536-0284  
**(After Hours: Please call 1-800-888-5533)**